Do Now · retrieval
Synoptic programming challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Give the Big O of a full scan of n values.

Q2

Which item does a queue remove first?

Q3

What does return count do?

Q4

Starting total=0, sum even values from [3,4,6].

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Inputs and output — Inputs are an array and a threshold; the result is the number of qualifying items.

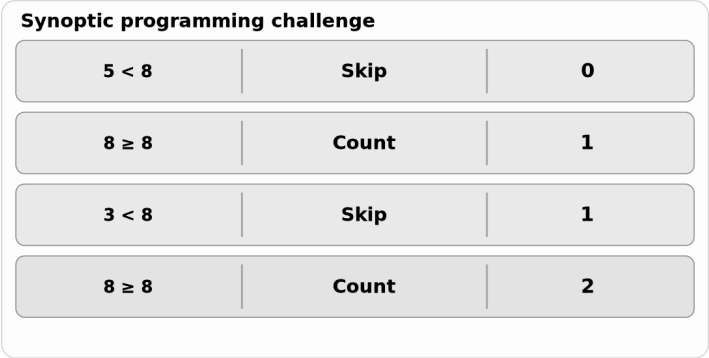
Core idea 2

Check each element — The counter increases only for values greater than or equal to the threshold.

Core idea 3

Test a boundary — A value equal to the threshold must count. For 5,8,3,8 at threshold 8, result is 2.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

decomposition — Break a problem into smaller tasks.

threshold — Value used as a cutoff.

counter — Variable that records how often an event occurs.

boundary test — Input at or next to a permitted limit.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Count values at least 7 in [4,7,9].

Follow the worked method

1. Predict — Skip 4 because it is below 7.
2. Explain — Count 7 and 9: the result is 2.

Source model · final illustrated step

Count values at least 7 in [4,7,9].

4 < 7	Skip	0
7 ≥ 7	Count	1
9 ≥ 7	Count	2

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

The first 6 counts and 2 does not.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Find CountAtLeast([2,5,7],5).

Question 2

Change the method so it counts values strictly greater than the threshold.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Give normal, boundary and erroneous test cases if the threshold must be non-negative.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The test score>=t excludes values equal to t.

Use key vocabulary from memory

1. Explain decomposition in your own words.

2. Explain threshold in your own words.

3. Explain counter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.